THE UNSPOOLER ASSEMBLY MANUAL

Step by step assembly instruction to help you create your UnSpooler.

Big Dog Custom Creations

Rev. 1 9/7/2025

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STEP 1 … Getting Started

You will need the following hardware to assemble the UnSpooler.

- 1 - BRINGSMART DC 12V 80rpm DC Worm Gear Motor 15kg.cm Self-Locking Reversed Engine Mini Turbine Geared (12V 80rpm)
- 1 - Bringsmart CCM6N DC PWM Motor Speed Controller 6V-30V Variable Speed Drive 6A Positive Inversion Mini Switch Controller (Speed Controller)
- 1 - DC-099 DC Female Panel Mount Charging Wire Adapter 5.5 x 2.5 MM Power Jack Cord Socket Threaded Female Mount Connector
- DC 12V 5A Power Supply Adapter Converter Transformer AC 100-240V Input with 5.5x2.5mm DC Output
- 1 - Polyurethane Coated Conveyor Roller 2" Diameter 10" Length
- 1 – 8mm to 12mm Rigid Motor Shaft Coupling Coupler Sleeve Coupling Adapter
- 4 - screw on rubber feet. You can choose the size and style of these feet.
- 11 - 6901-2RS Bearings, 12mm x 24mm x 6mm Bearings
- 2 - F695ZZ Flanged Ball Bearing, 5mm x 13mm x 4mm Miniature Bearing
- 2 - 12mm x 300mm (0.472 x 11.81 inches) Vigorous Linear Motion Rods
- 1 - 6063 Aluminum Round Tube, 300mm Length 12mm OD 5mm Inner Dia Seamless Aluminum Straight Tubing
- 1 - Pkg. M3x4.0mm, OD4.6mm Heat-Set Threaded Inserts
- 1 - Pkg. M4 x 5 OD x 4 L Heat-Set Threaded Inserts
- 2 – M6x20mm Button Head Screws
- 2 – M6x16mm Socket Head Screw
- 4 – M5x25mm Socket Head Screws
- 4 – M5x12 or 16mm Button Head Screws, (depends on the feet you use).
- 4 – M4x75mm Socket Head Screws
- 4 – M4x45 Button Head Screws
- 4 – M4x40mm Socket Head Screws
- 4 – M4x16mm Socket Head or Button Head Screws
- 5 – M4x8mm Socket Head Screw
- 8 – M4x8mm Button Head Screws
- 4 – M4 Washers
- 8 – M3x16mm Socket Head screws
- 4 – M3x16 Button Head Screws
- 2 – M3x8mm Socket Head Screw
- 6 – M3x6mm Socket Head Screw
- 2 - 6x1/2in. flat head wood screws

STEP 2 … Heat Inserts

Printed Parts Needed for this Step:

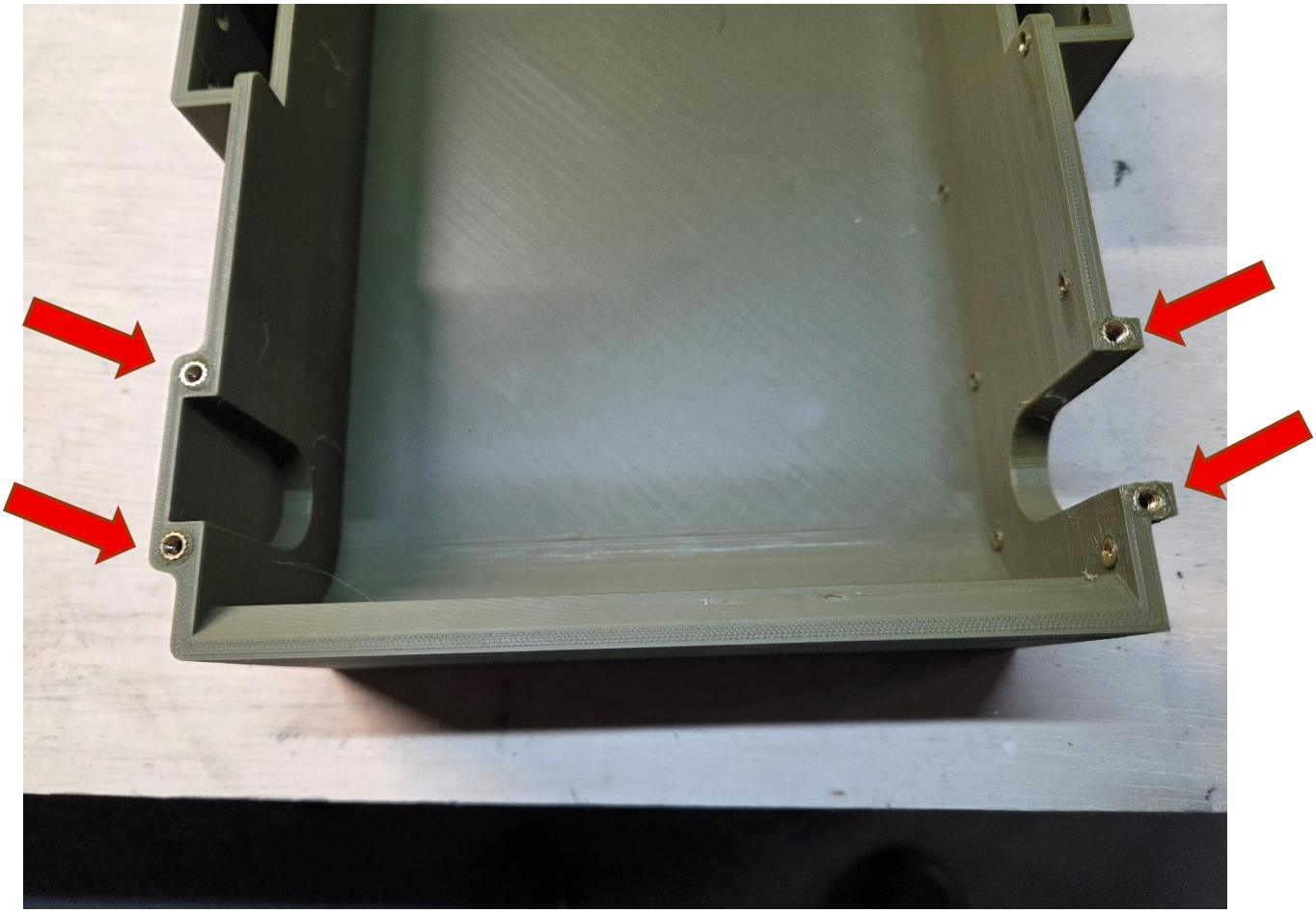
- Main Body (revision date)
- Printed Speed Upright (revision date)

Hardware Needed for this Step:

- 22 - M3 x 4OD x 4.6L Brass Heat Inserts
- 4 - M4 x 5OD x 4L Brass Heat Inserts

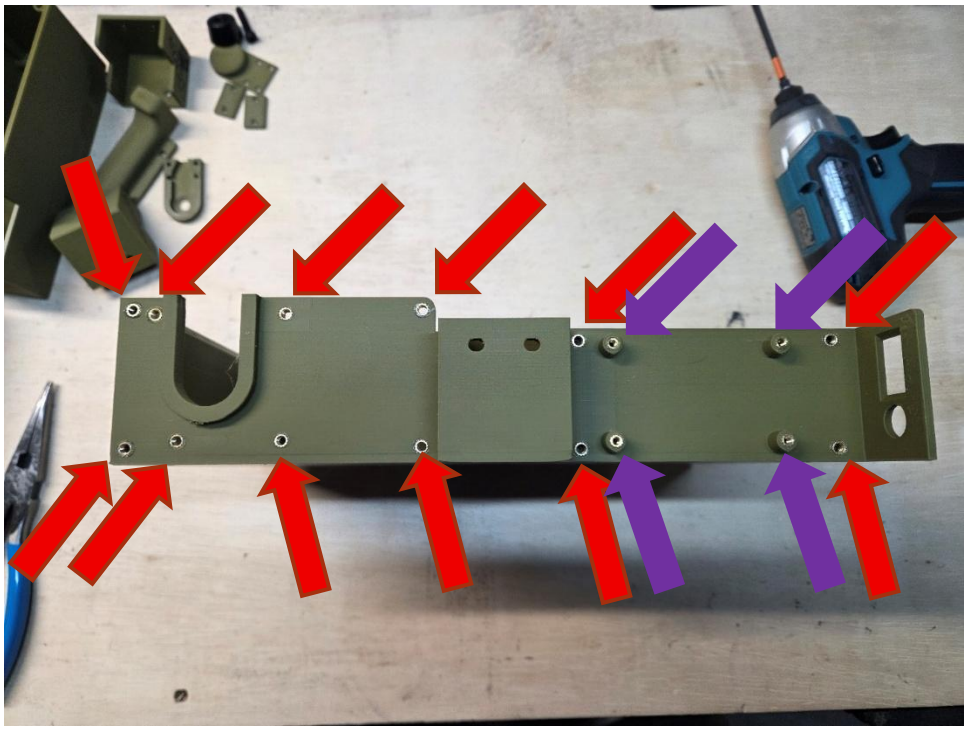
Insert the brass heat inserts into these locations in the sizes indicated below.

M4 Heat insert locations



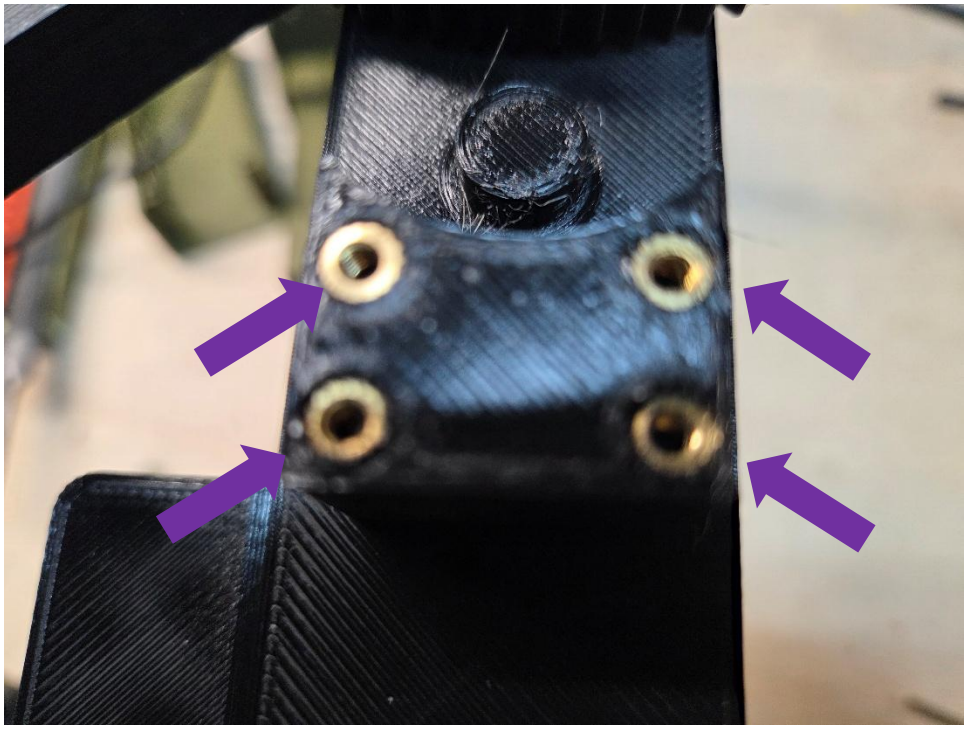
M4 Heat insert locations

M3 Heat insert locations



Printed Speed Arm Upright. After installing heat inserts, set this part aside till Step 7.

M3 Heat insert locations



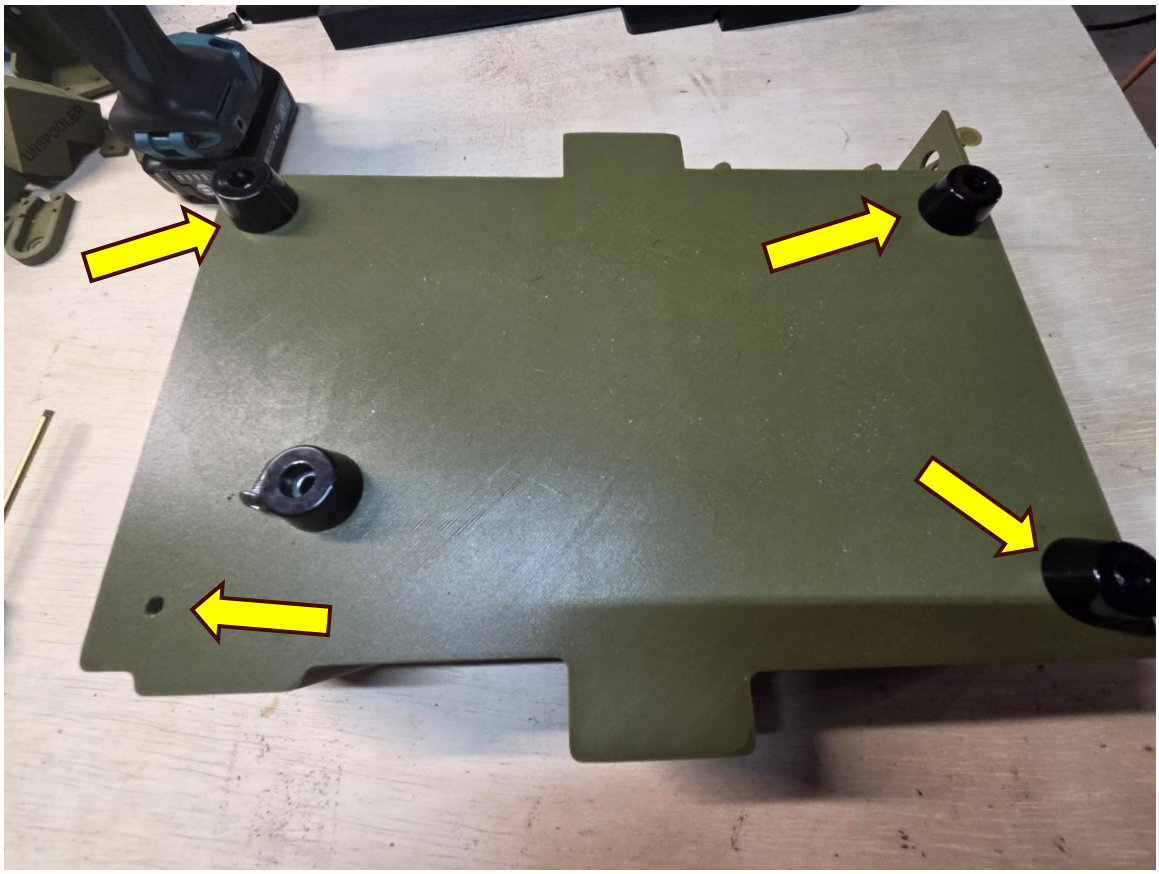
STEP 3 … Attaching the feet

Printed Parts Needed for this Step:

- Main Body (revision date)

Hardware Needed for this Step:

- 4 – screw on rubber feet. You can choose the size and style of these feet. They will be attached with M5x12 Button Head Screws.
- 4 – M5x12 or 16mm, Button Head Screws. (depend on the feet you use)



Attach the feet into the screw holes on the bottom of the **Main Body**. You want the screws to go about 4 to 6mm into the screw hole. You may need to choose a different length screw depending on the feet you chose.

STEP 4 … The Main Body – The Idle Roller Set Up

Printed Parts Needed for this Step:

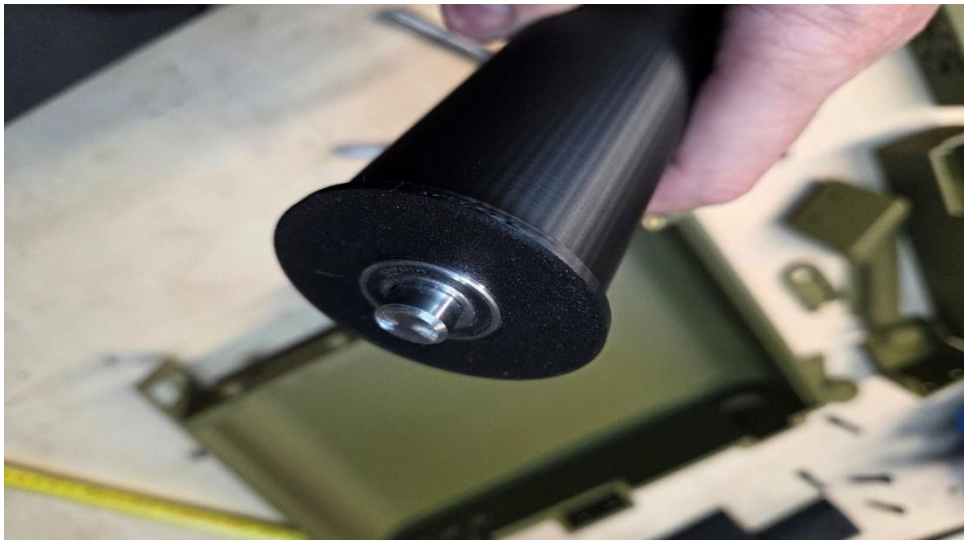
- Idle Roller (revision date)

Hardware Needed for this Step:

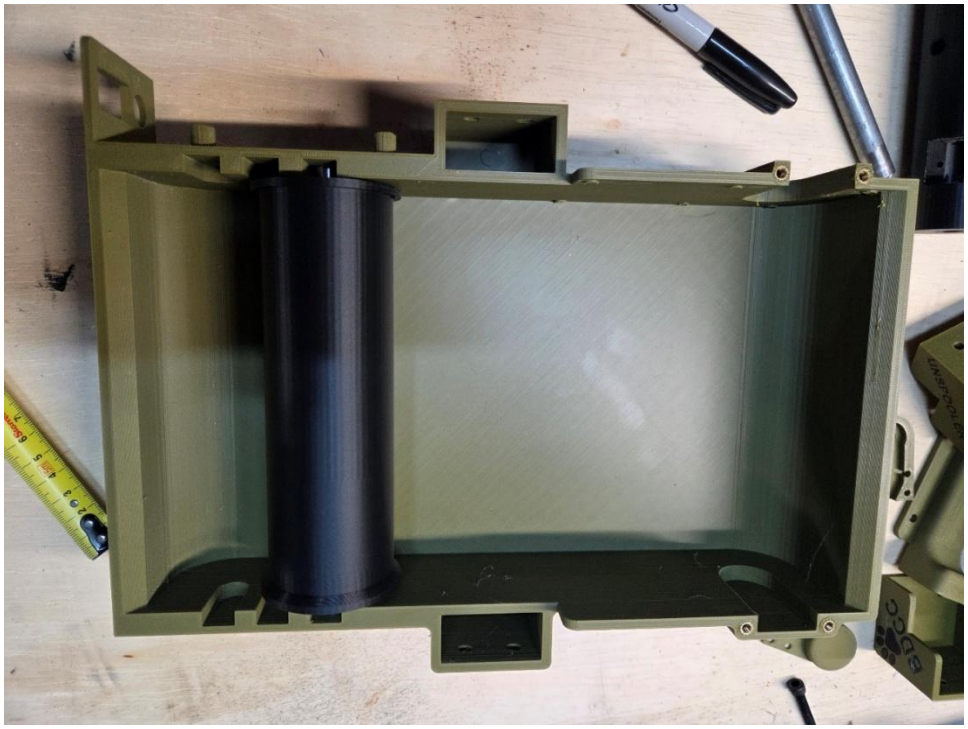
- 2 - 6901-2RS Deep Groove Bearings, ID 12mm x OD 24mm x Width 6mm Double Rubber Sealed Ball Bearings
- 1 - 12mm x 300mm (0.472 x 11.81 inches) Vigorous Linear Motion Rods, (cut to 172mm)



Cut the 12mm shaft to 172mm in length. These shafts have a hardened outer casing, so it is better to use an abrasive cutter like a chop saw or a 4” grinder with a cutoff wheel on it. Clean off any burs that may be on the cut side.



Then, slide on the printed part **Idler Roller** onto the shaft. Place a ID 12mm x OD 24mm x Width 6mm bearing on both ends to the shaft. Insert the roller assembly into the slots at the back of the **Main Body**. (below)



STEP 5 … The Drive Roller Set Up

Printed Parts Needed for this Step:

- 1 - Drive Roller Center (revision date)
- 2 - Drive Roll Cap (revision date)
- 2 - 2mm Spacer (revision date)
- 1 - Left Front Hold Down (revision date)
- 1 - Right Front Hold Down (revision date)

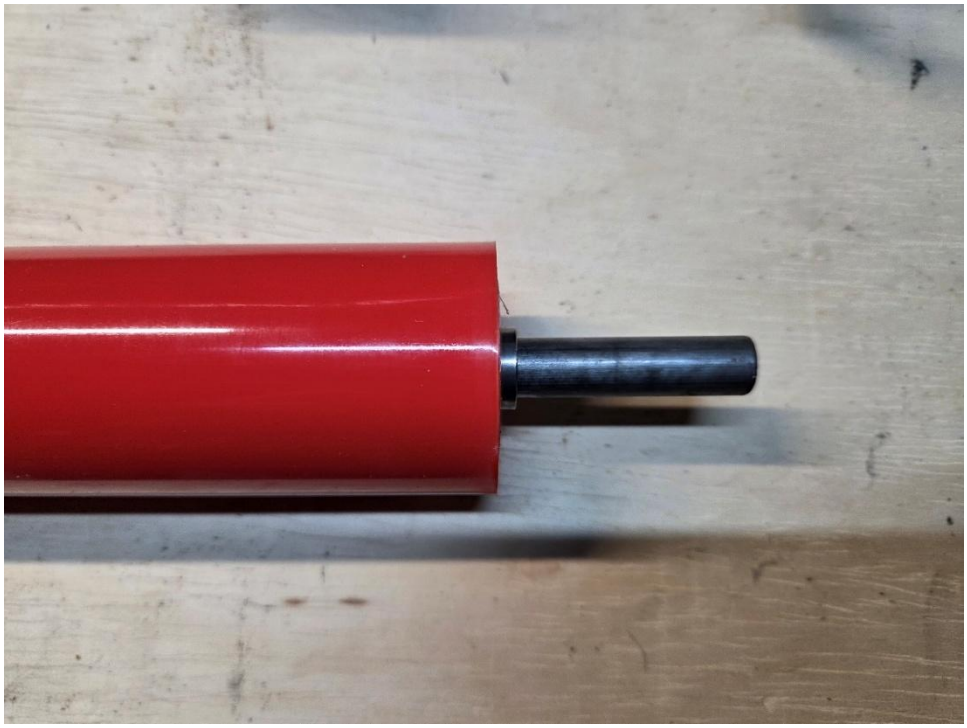
Hardware Needed for this Step:

- 2 - 6901-2RS Deep Groove Bearings, ID 12mm x OD 24mm x Width 6mm Double Rubber Sealed Ball Bearings
- 1 - Polyurethane Coated Conveyor Roller 2" Diameter 10" Length
- 1 – 8mm to 12mm Rigid Motor Shaft Coupling Coupler Sleeve Coupling Adapter
- 4 – M3x16 socket head cap screws
- 2 - 6x1/2in. flat head wood screws
- 4 – M4x8mm Button Head Screws
- 1 – M4x8mm Socket Head Cap Screw

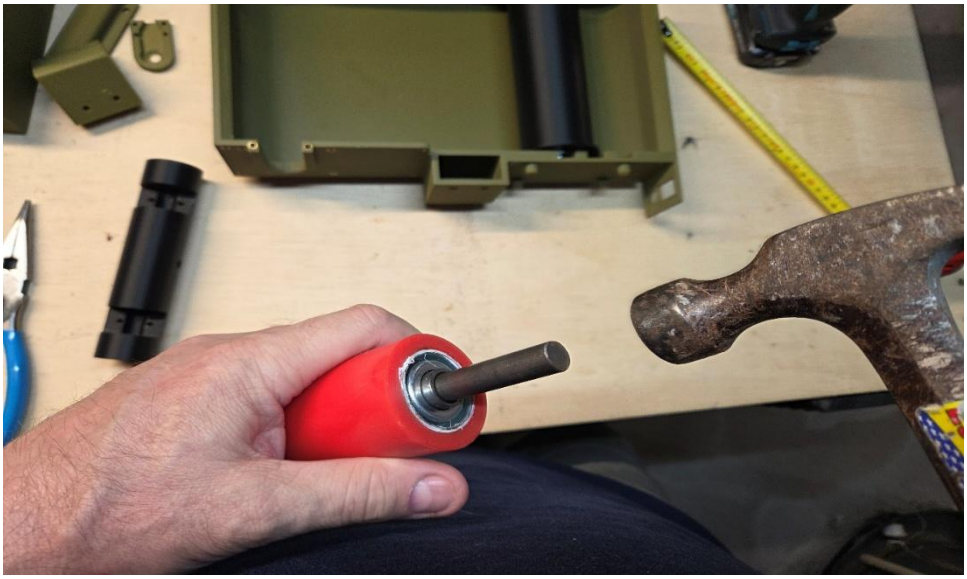
First, we are going to prepare the polyurethane roller. It needs to be unassembled.



One end of the shaft will pull out. Pull out that end of the shaft and tilt it to one side so it doesn't pop back out the other end.



It will look like this. Then hit it with a hammer till the bearing cap on the other end comes out of the roller.



Using the shaft again, use it to knock out the other bearing cap in the same way. When the other end is out, you should have several pieces that look like this.



You can discard the two bearing caps and the spring. They are not going to be used. Clean the burrs off the shaft as we will be using it in this step. Leave the tabs on, they will help keep the shaft from spinning in the roller assembly.

Cut the shaft down to 190mm in length and clean up the cut end.



Set up the center pieces of the drive roll as follows. Put the 8mm x 12mm shaft coupler on the end of the shaft you cut down and tighten the screws. Slide the shaft assembly through the center of the printed part **Drive Roller Center**. Then assemble the remaining components as follows.

Left to right:

- ID 12mm x OD 24mm x Width 6mm Double Rubber Sealed Ball Bearing
- 2mm printed spacer
- Printed part **Drive Roller Center**
- 2mm printed spacer
- ID 12mm x OD 24mm x Width 6mm Double Rubber Sealed Ball Bearing
- 8mm to 12mm Sleeve Coupling Adapter



The shaft should be flush with the outside of the bearing on the left side.

Insert the printed part **Drive Roll Cap** into the 2 open spaces. Attach with 4 - M3x16 Socket Head Cap Screws. Tighten till just past snug.



CAUTION DO NOT OVERTIGHTEN.

There is a hole in the center on the printed part **Drive Roller Center**. Put a M4x8 Socket Head Cap screw in this opening and tighten till just past snug.

Now take the roller that you removed the center parts from and cut it to size. I recommend that you cut 10mm off one side first. This removes the any bent or damaged places. Then cut the roll to 160 mm in length having cut off both ends. This ensures a better fit. A band saw or chop saw works best for this. This is what you should have to work with now.



Take the cut section of the roller and using an 8mm drill bit, drill through the rubber portion of the roller to expose the metal underneath. DO NOT drill through the roller, just remove a small piece of the rubber. Then using a 3mm drill bit, drill through the roller. Repeat this on the opposite side of the roller. It should look like the image below.



The image below shows what you should have ready at this point.



Insert the drive roller center assembly into the roller sleeve. Take the 2 #6x1/2” Flat Head Wood Screws and insert them into the two drilled holes. You now have a completed roller assembly.



Put the completed assembly into the **Main Body**. The assembly should be flush with the non-driven side of the Main Body openings. Make sure all the components on the shaft are pushed to the non-driven side. There should be a small gap between the driven side bearing and the shaft coupling. Insert the printed parts Left Front Hold Down and Right Front Hold Down into the openings. The one with the rounded corners goes into the non-driven side.



Insert g the 4 – M4x8 Button Head Screws into the hold downs and tighten. The roller should spin freely without binding up.

STEP 6 … The Drive Motor Set Up

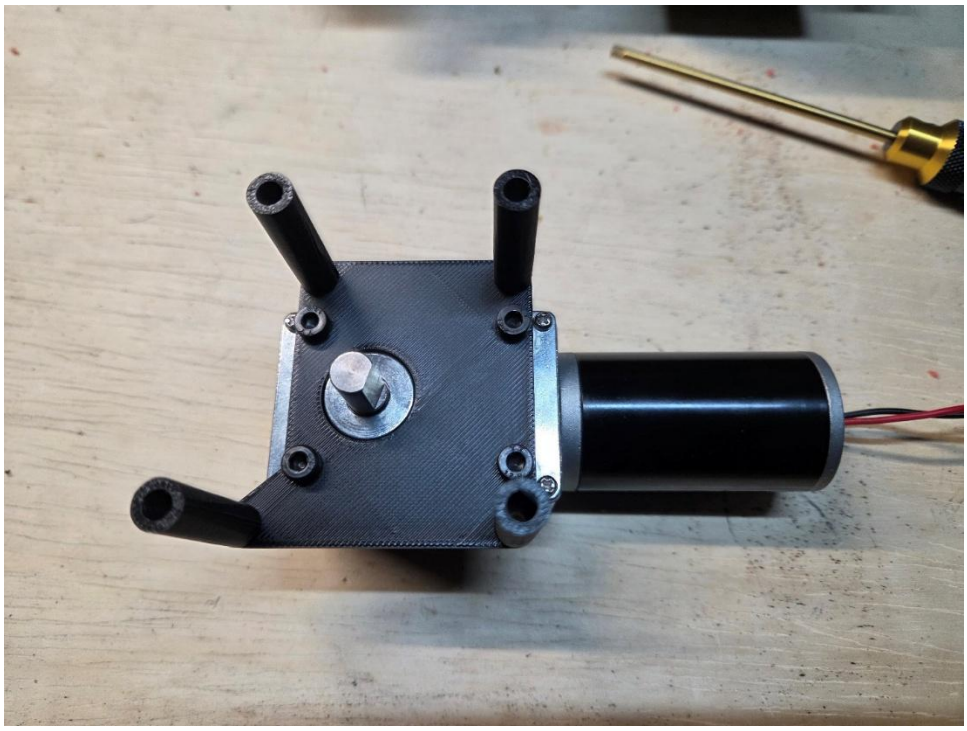
Printed Parts Needed for this Step:

- 1 - Motor Mount (revision date)

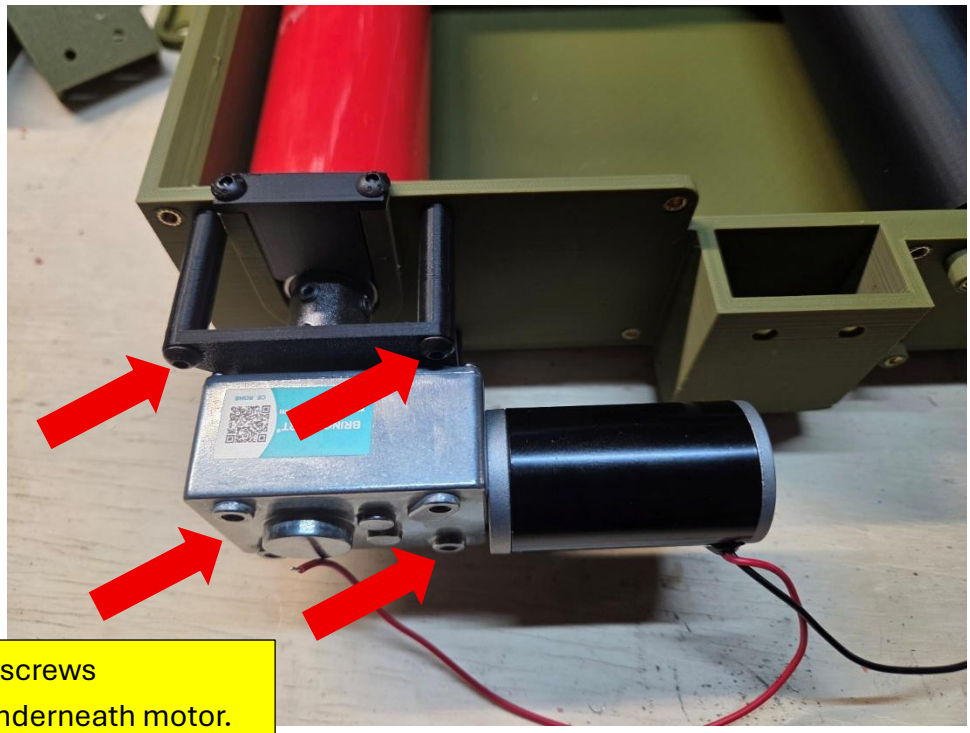
Hardware Needed for this Step:

- 1 - BRINGSMART DC 12V 80rpm DC Worm Gear Motor 15kg.cm Self-Locking Reversed Engine Mini Turbine Geared Motor
- 4 – M4x8mm Socket Head Screws
- 4 – M4x40mm Socket Head Screws

Attach the **Motor Mount** to the motor using 4 – M4x8mm Cap Screws. Notice the orientation of the motor mount to the motor for proper placement. Tighten screws.



Now place the motor assembly onto the side of the Main Body. Insert 4 – M4x40mm Socket Head or Button Head Screws and tighten. Spin the roller till the flat spot on the motor shaft is aligned with the set screws and tighten.



2 screws underneath motor.

The **Motor Cover** will be added in a later Step.

STEP 7 … The Uprights and Overhead Bar Set Up

Printed Parts Needed for this Step:

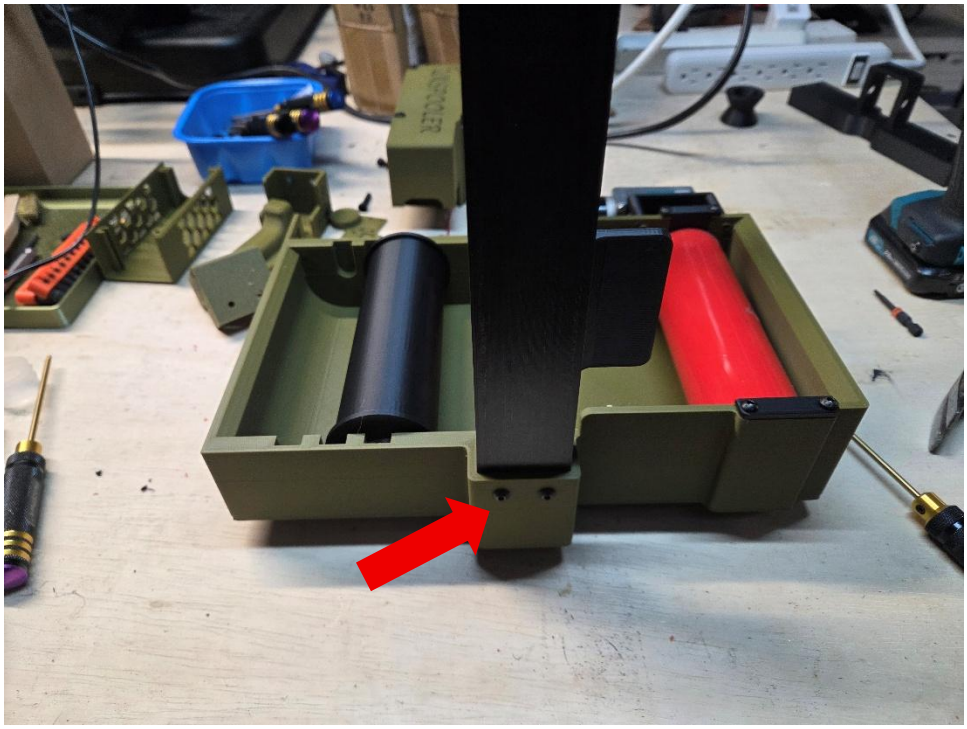
- 1 – Printed Idler Upright (revision date)
- 1 - Printed Speed Upright (revision date)
- 1 - Idler Arm (revision date)
- 1 – Speed Arm Gear 72 teeth (revision date)
- 1 - Overhead Rail (revision date)
- 1 - 39mm Roller (revision date)
- 2 - Roller Shaft Cap (revision date)
- 1 – Centering Roller (revision date)
- 2 – 21mm Spacer (revision date)

Hardware Needed for this Step:

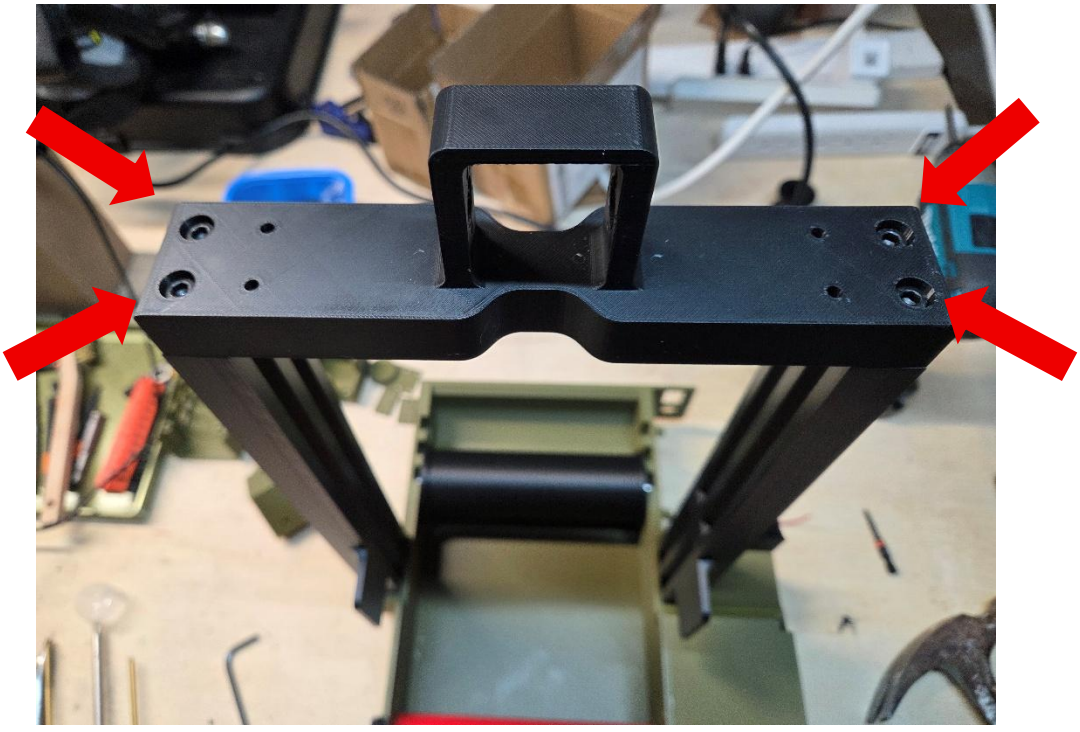
- 2 - 6901-2RS Deep Groove Bearings, ID 12mm x OD 24mm x Width 6mm Double Rubber Sealed Ball Bearings
- 2 - F606ZZ Flanged Ball Bearing 6x17x6mm Chrome Steel Double Shield
- 1 - 6063 Aluminum Round Tube, 300mm Length 12mm OD 5mm Inner Dia Seamless Aluminum Straight Tubing
- 4 – M4x16mm Button Head Screws
- 4 – M4 Washers
- 4 – M5x25mm Cap Screws
- 2 – M6x16mm Cap Screw
- 2 – M3x8mm Cap Screw
- 2 – M6x20mm Button Head Screws
- 4 - M3 x 4OD x 4.6L Brass Heat Inserts
- 1 - The piece of the shaft that was cut off in STEP 4 from the Idler Roller, cut to 46mm

Insert **Idler Arm** into the opening and attach with 2 – M4x16mm Socket Head or Button Head Screws.

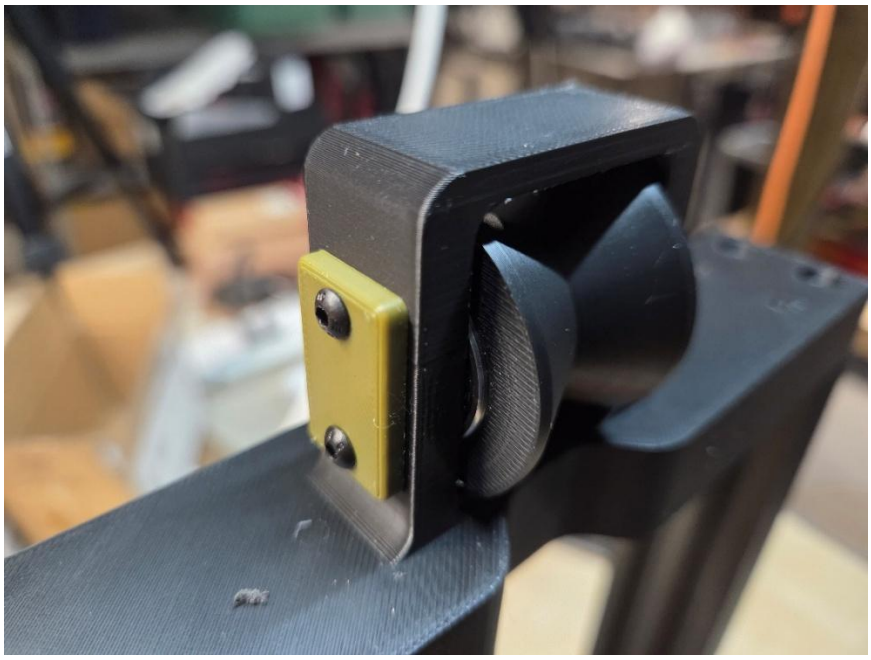
Repeat this for the other side. The **Printed Speed Upright**.



Place the Overhead Rail on top of the two upright arms and attach with 4 – M5x25mm Cap Screws.



Insert 2 - ID 12mm x OD 24mm x Width 6mm bearings into the 39mm Roller. These will be a tight fit. Insert the 39mm Roller assembly into the top of the **Overhead Rail** and insert the 46mm cut piece of shaft through the center. Install the 2 - **Roller Shaft Cap** with 4 – M3x8mm Button Head Screws.



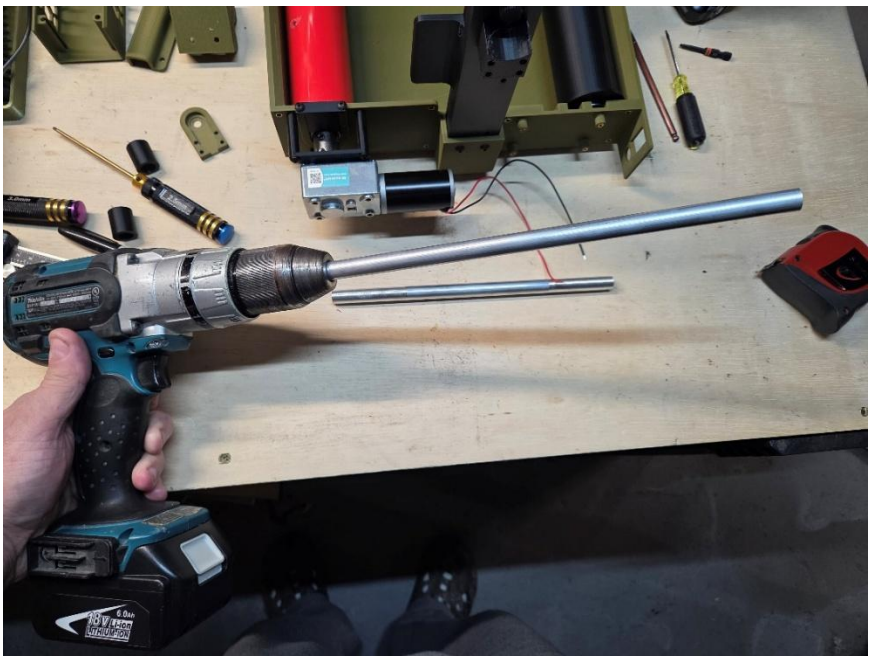
Take the Idler Arm and insert an 12mm x OD 24mm x Width 6mm bearing on the inside and a Flanged Ball Bearing 6x17x6mm on the outside. Attach to the non-driven side of the uprights with a M6x16mm Cap Screw.



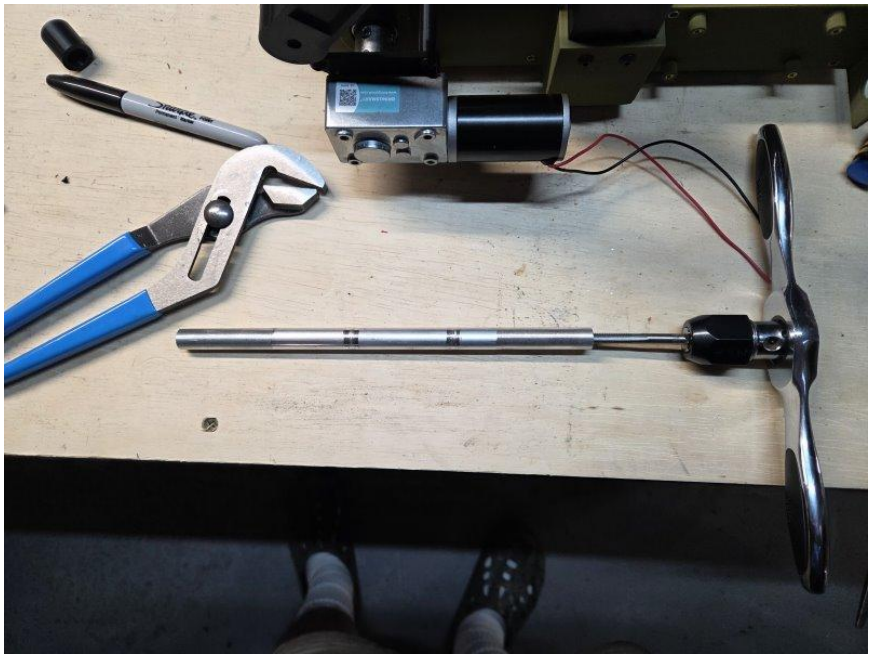
Next, take the Speed Arm Gear and insert a 12mm x OD 24mm x Width 6mm bearing on the inside and a Flanged Ball Bearing 6x17x6mm on the outside. Attach to the driven side of the uprights with a M6x16mm Cap Screw.



Next is the centering roller in the front. You will need the 12mm aluminum round tube. Cut the tube down to 226mm. Clean any burs off the cut end. The tube will be a little too large for the bearings. You will need to sand the tubing down a little till they fit. See the next image to see how I accomplish this. I put the tube in a drill like a bit and use sandpaper while spinning the drill. This will remove material evenly. Do this for both ends until the bearings fit properly.



Once the bearings fit on the shaft it is time to tap the ends with a M6x1.0 tap. This is a soft metal so be sure to use oil on the tap and clean it off regularly. The threads can be damaged by a dirty tap. Use a wire brush to clean the threads of the tap.



Once both ends are tapped, assemble the centering roller as shown.

Shaft in center, 21mm spacer (printed part), 12mm x 24mm x 6mm bearing, centering roller (printed part), 12mm x 24mm x 6mm bearing, 21mm spacer (printed part).



Mount the Centering Roller in the arms of the Unspooler and thread in the 2-M6x20mm Button Head screws in the ends. Tighten till snug but make sure the roller still turns. Pay attention to the orientation of the centering roll or the filament will work its way off the end.



STEP 8 … The Speed Controller Set Up

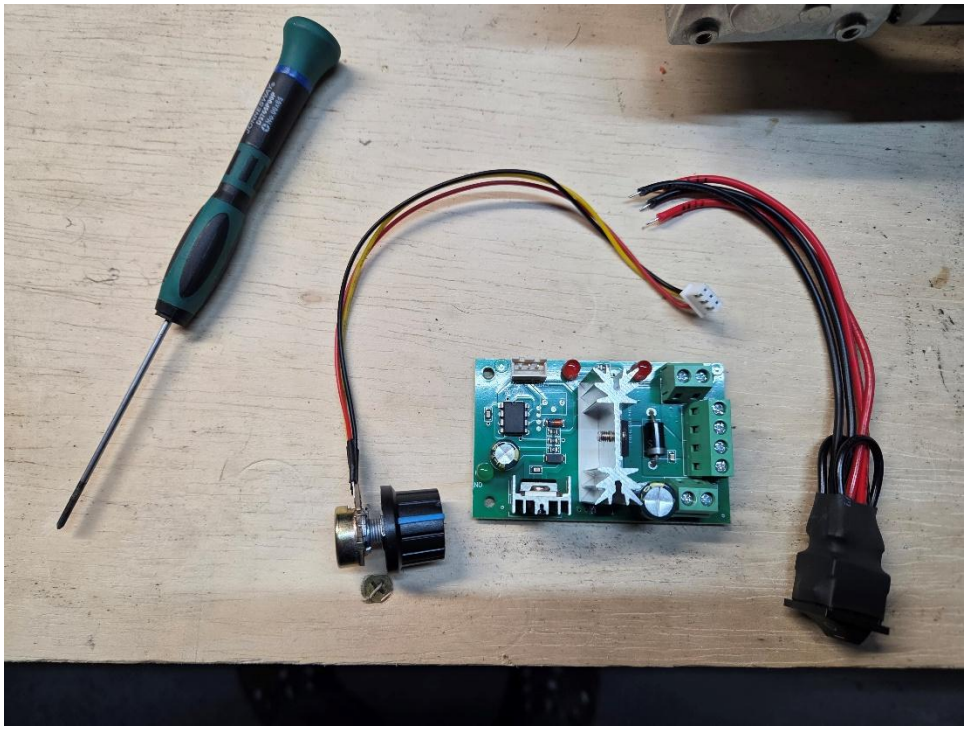
Printed Parts Needed for this Step:

- 1 - Spur Gear (24 teeth) (revision date)
- 1 - Potentiometer Bracket (revision date)
- 1 - Pot Bracket Cover (revision date)
- 1 – Side Cable Cover (revision date)
- 1 – Motor Cover (revision date)
- 1 – Motor Controller Cover (revision date)

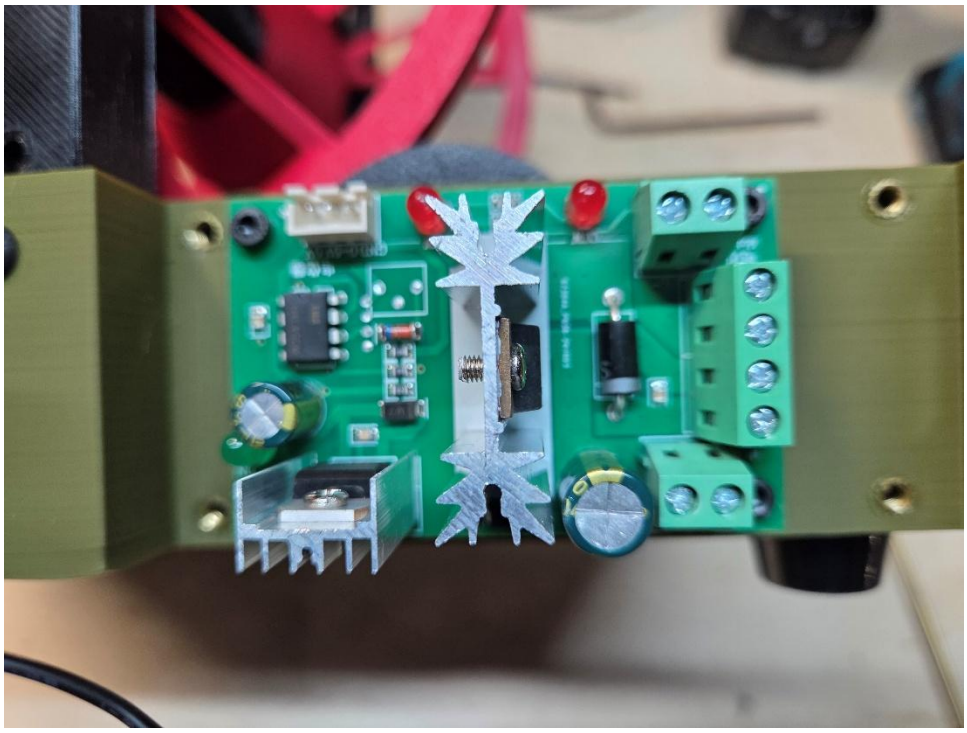
Hardware Needed for this Step:

- 1 - Bringsmart CCM6N DC PWM Motor Speed Controller 6V-30V Variable Speed Drive 6A Positive Inversion Mini Switch Controller (Speed Controller)
- 1 - DC-099 DC Female Panel Mount Charging Wire Adapter 5.5 x 2.5 MM Power Jack Cord Socket Threaded Female Mount Connector
- DC 12V 5A Power Supply Adapter Converter Transformer AC 100-240V Input with 5.5x2.5mm DC Output
- 1 - 6901-2RS Deep Groove Bearings, ID 12mm x OD 24mm x Width 6mm Double Rubber Sealed Ball Bearings
- 4 – M4x75mm Cap Screws
- 4 – M4x45 Button Head Screws
- 4 – M4x8mm Button Head Screws
- 4 – M3x16 Button Head Screws
- 4 – M3x6 Socket Head Screws

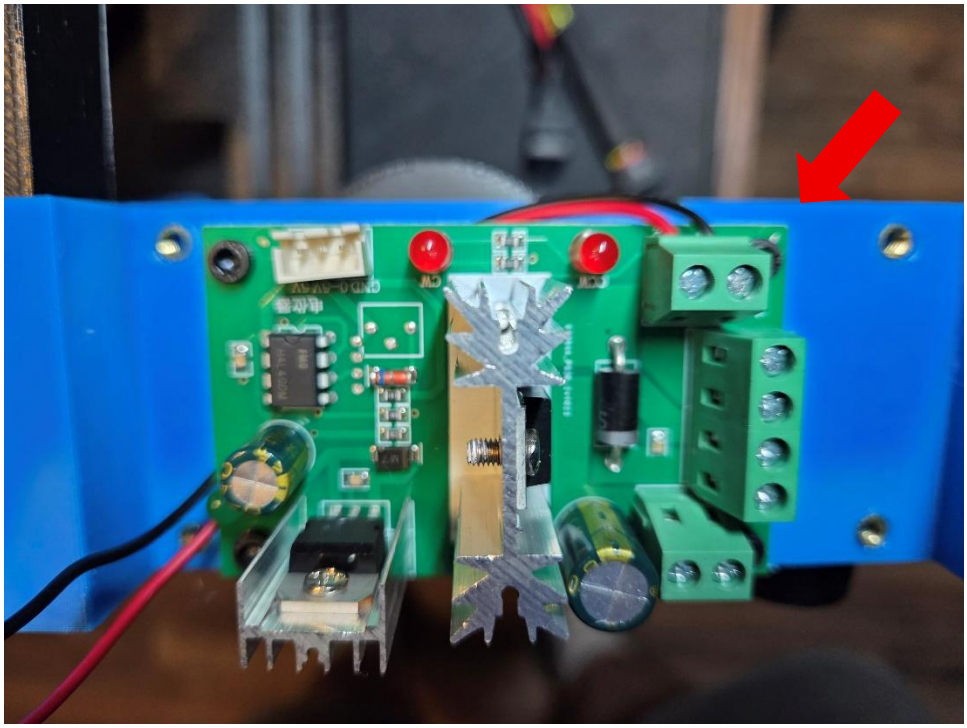
Remove the speed controller board from package. Disconnect the switch and the potentiometer from the board. *Be sure to label the wires on the switch to reconnect them later*.



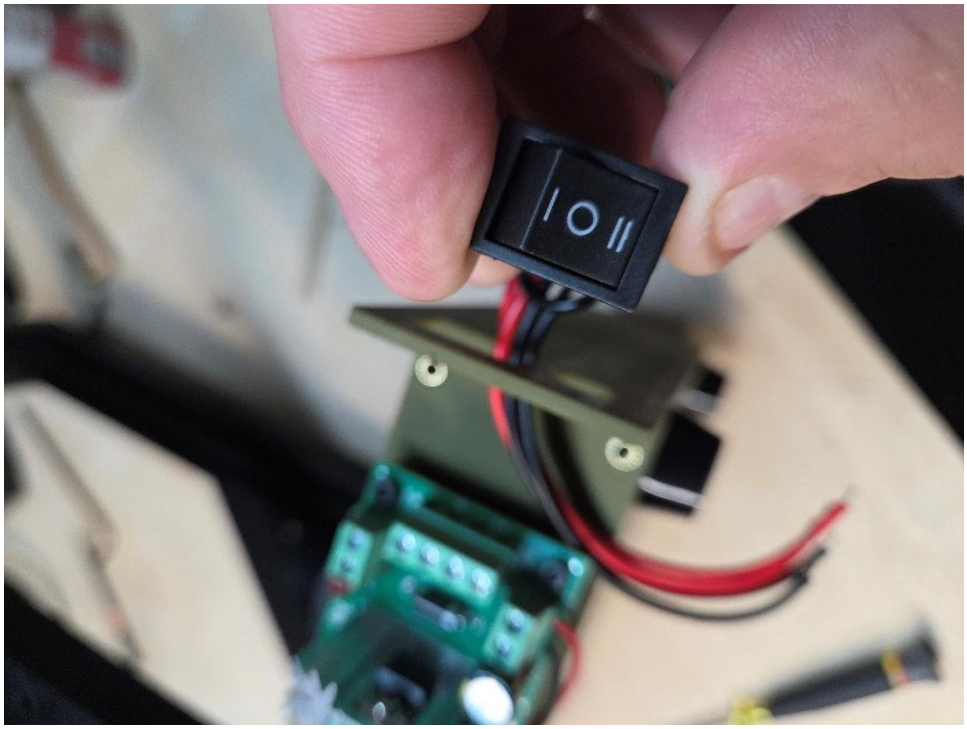
Attach the board to the Unspooler with 4 – M3x6mm socket head screws. The socket for the potentiometer should be on top as shown.



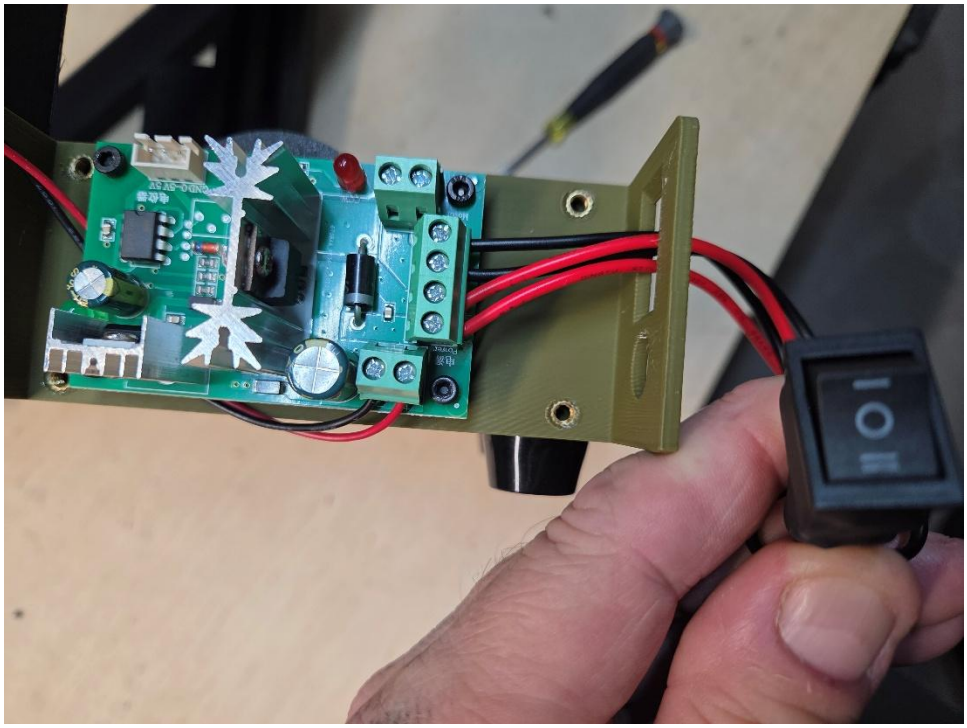
Attach the motor wires to the board, Red to the M2 and Black to M1 will be on the top of the board. I like to tuck the wires under the board to keep them out of the way when installing the cover.



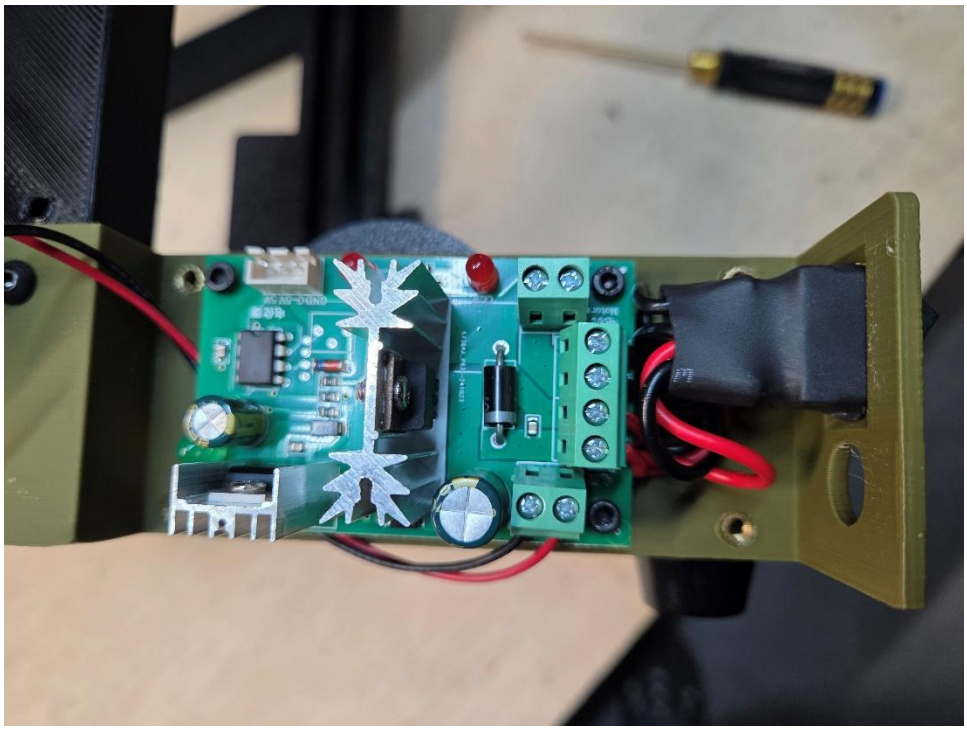
Take the switch and slide the wires through the opening at the end for the switch. The Number 1 should be on top when this is installed.



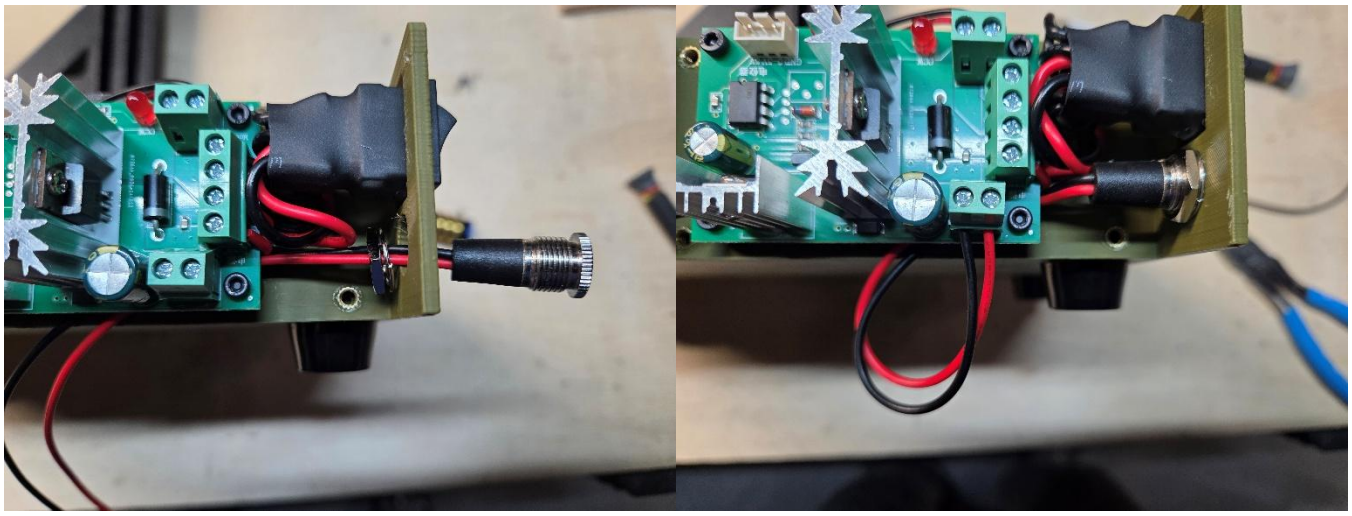
Start connecting the wires back on to the board while tucking the wires under the board. This will be a tight fit, but it can be done.



The switch will snap fit into the opening. Completed image.



Take the 5.5mm x2.5mm DC Socket adapter and push the wires through the back panel. Put the washer and the nut that was supplied on the wire as shown and install them. Snug down the nut, trim the wires to length leaving some extra, and tuck the wires under the board. Red goes to +Positive and Black goes to -Negative.



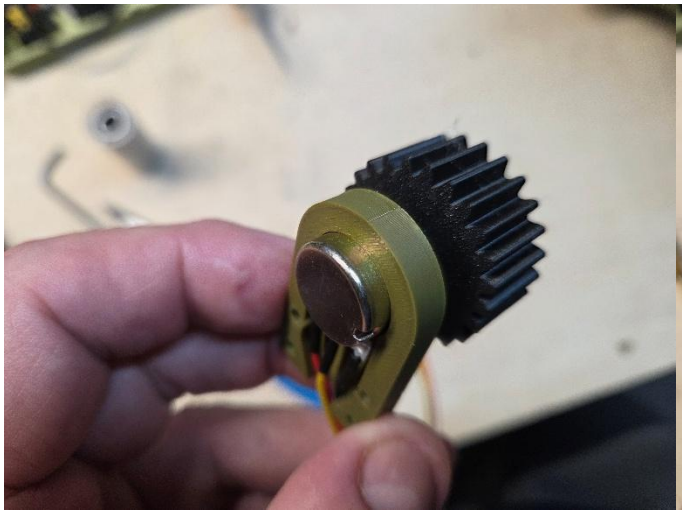
Next, take the potentiometer and the Potentiometer Bracket and push the potentiometer through the bracket. Install the nut on the back side. It should be a snug fit.



Take the 24-tooth gear and press a 6901-2RS Bearing inside of it.

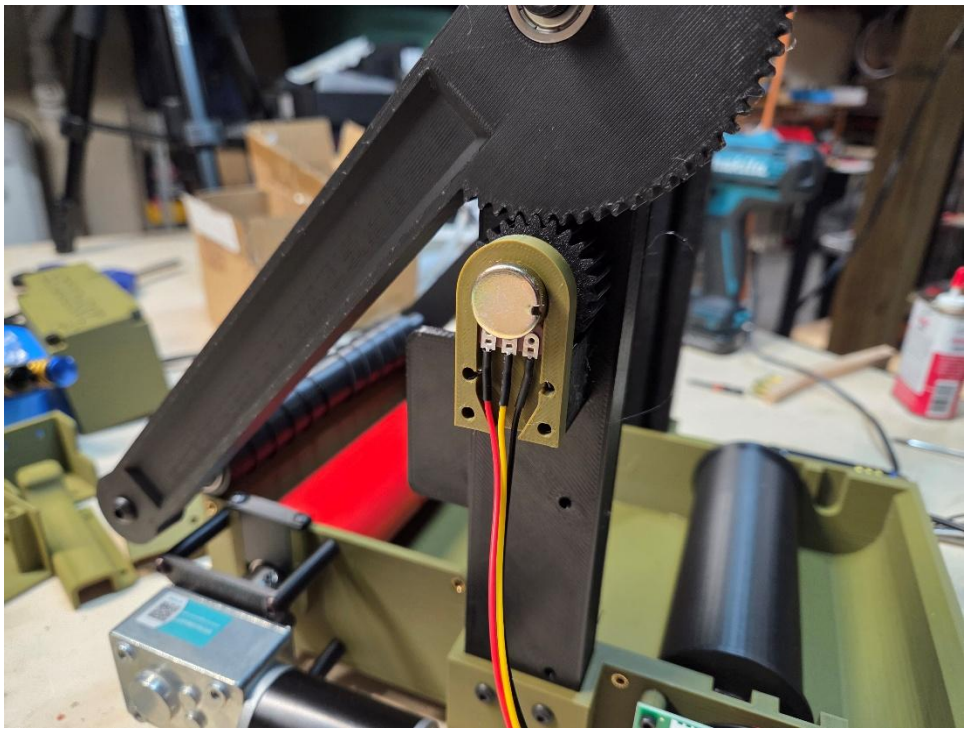


Take the potentiometer assembly and press the shaft through the flat side of the gear. The shaft should not protrude out the back by bearing. Plug the wire connector into the board. The wire should be just long enough for final assembly.

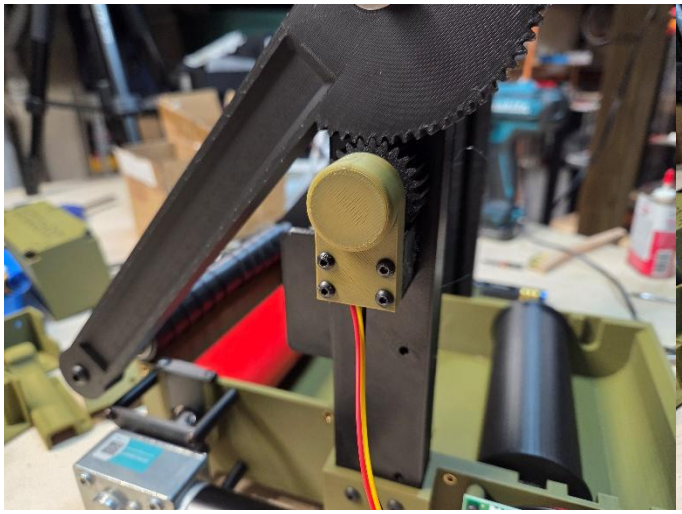


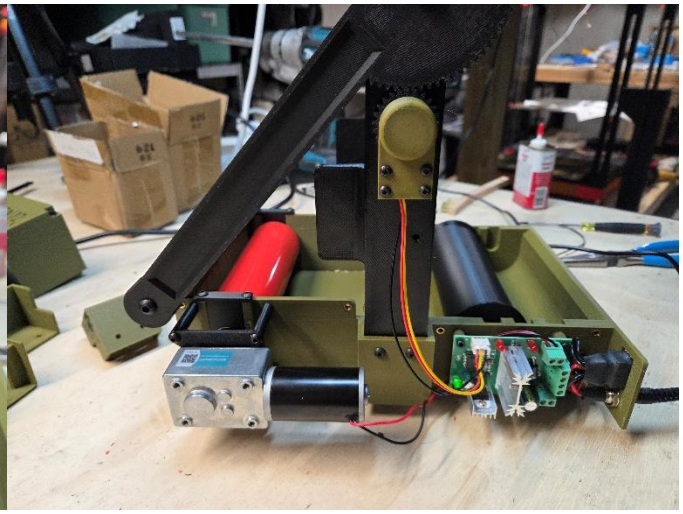


You should have a functional UnSpooler now. We need to adjust the potentiometer before installing it onto the speed arm. Plug the 12DC power supply into the jack on the back. Turn the 24-tooth gear and the front roller should spin. Turn the 24-tooth gear back until the front roller just stops. The front arm should be all the way down. Press the assembly onto the speed arm without moving the 24-tooth gear.

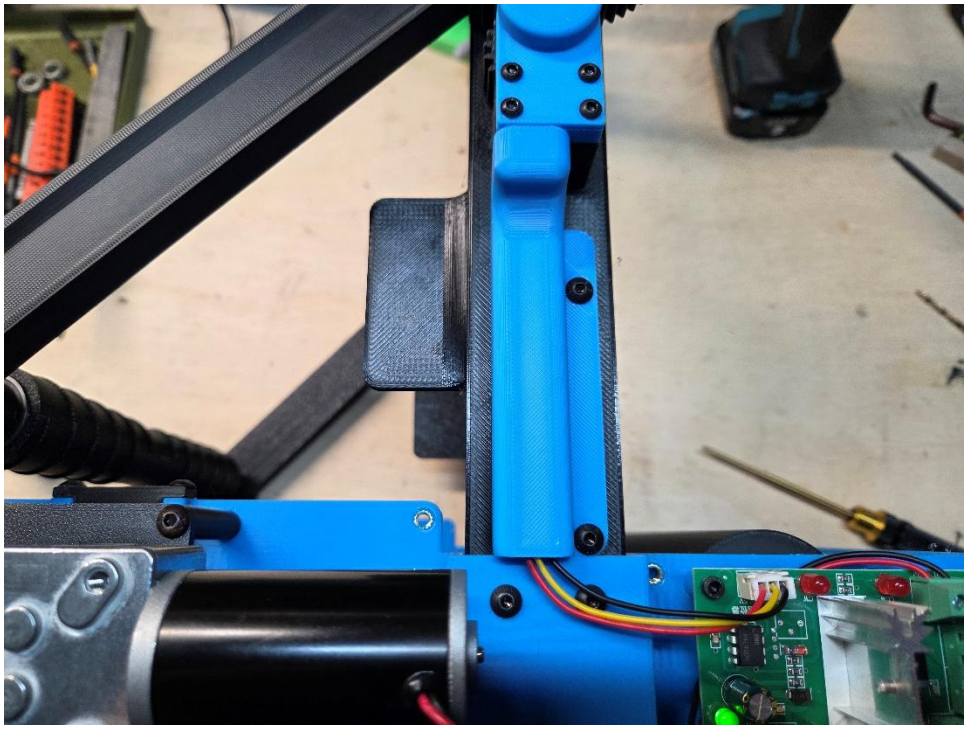


While holding the potentiometer assembly in place, lift the front arm. The front roller should start to rotate. The higher you raise the arm, the faster the front roller will spin. Lowering the front arm to the bottom should stop the roller. If not, then pull the assembly off and readjust till it does. When this is done, place the potentiometer bracket cover in place and insert 4-M3x16mm button head screws to hold it.

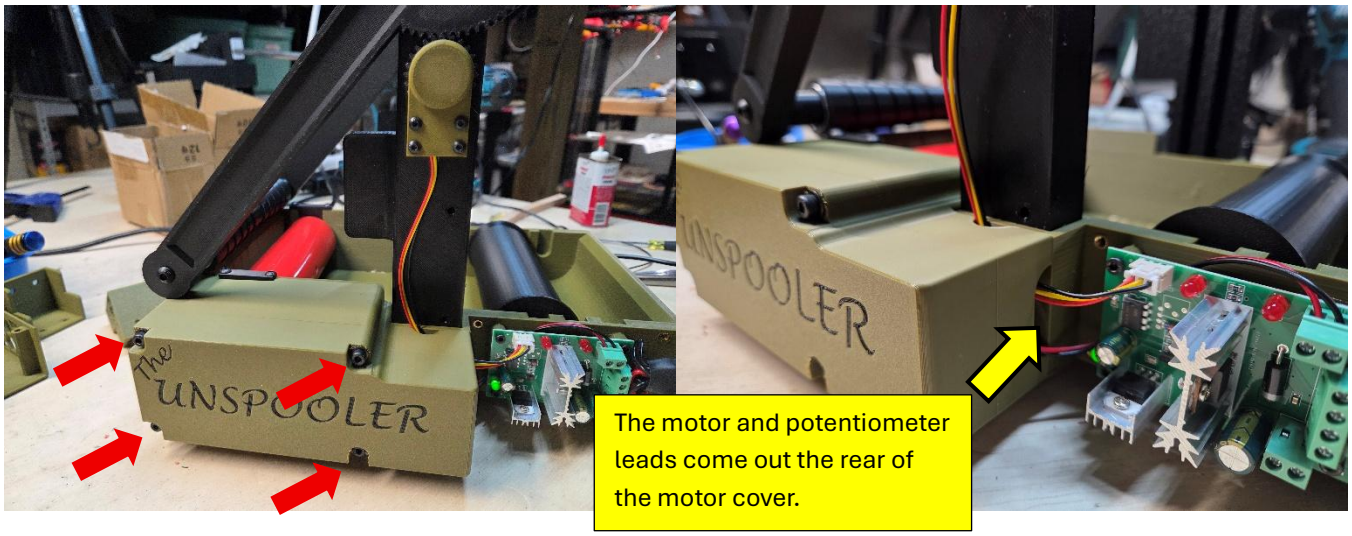




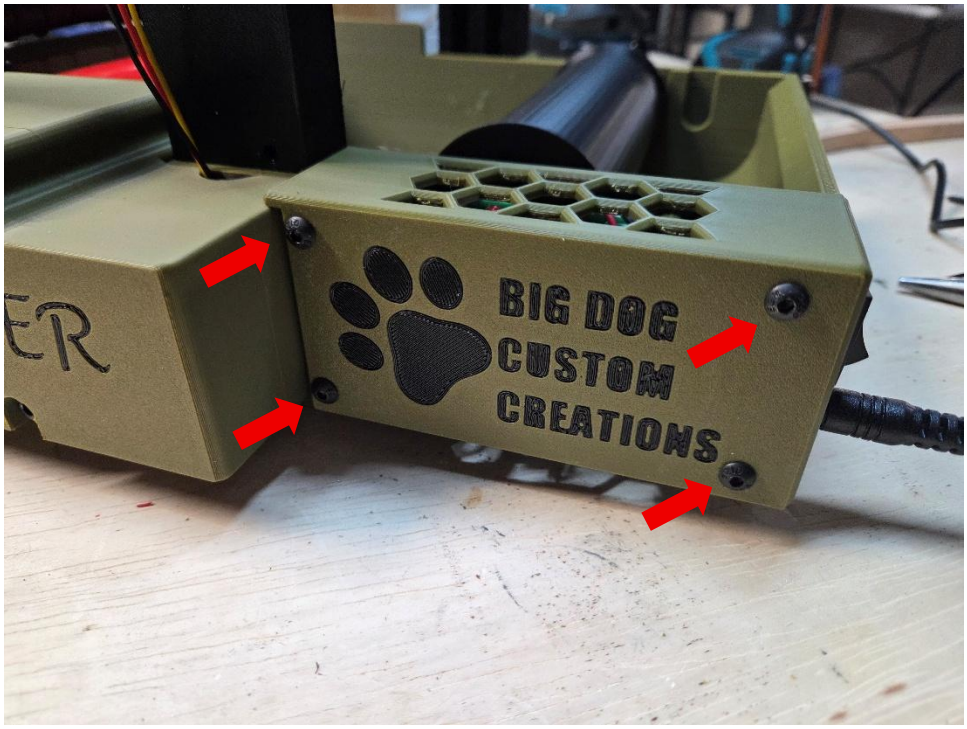
Mount the side cable cover next using 4 – M4x8mm Button Head Screws.



Mount the Motor Cover using 4 – M4x75mm Cap Screws with a M4 washer. As shown below. Moke sure the motor leads and potentiometer wires are sticking out the rear of the cover.



Mount the Motor Controller cover using 4 – M4x45 Button Head screws.



STEP 9 … Optional Items

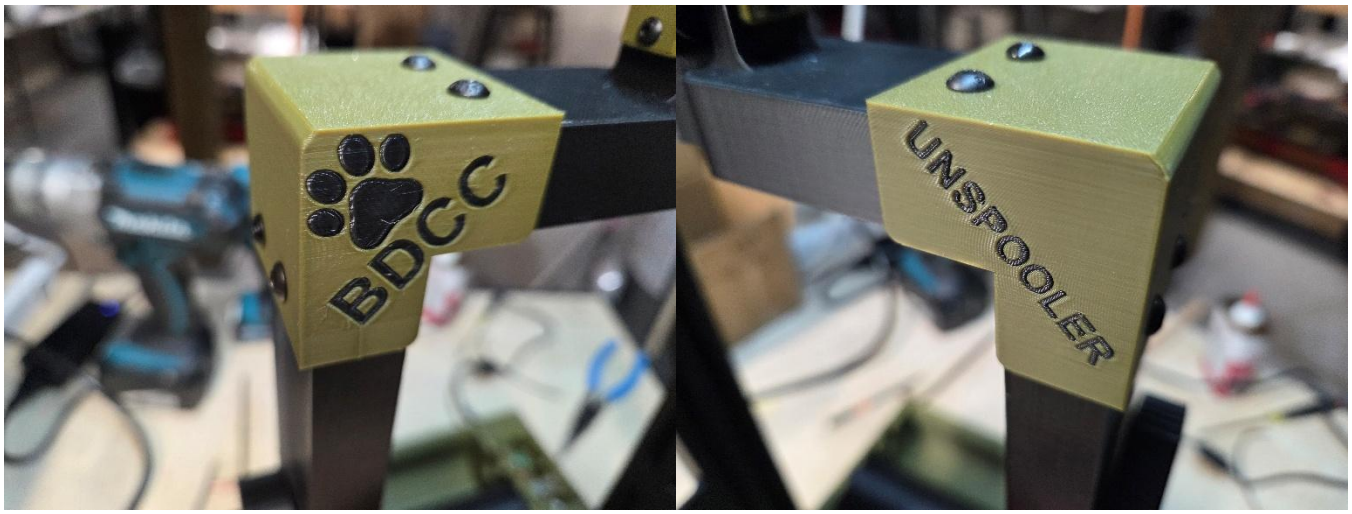
Printed Parts Needed for this Step:

- 2 – Corner Décor BDCC

Hardware Needed for this Step:

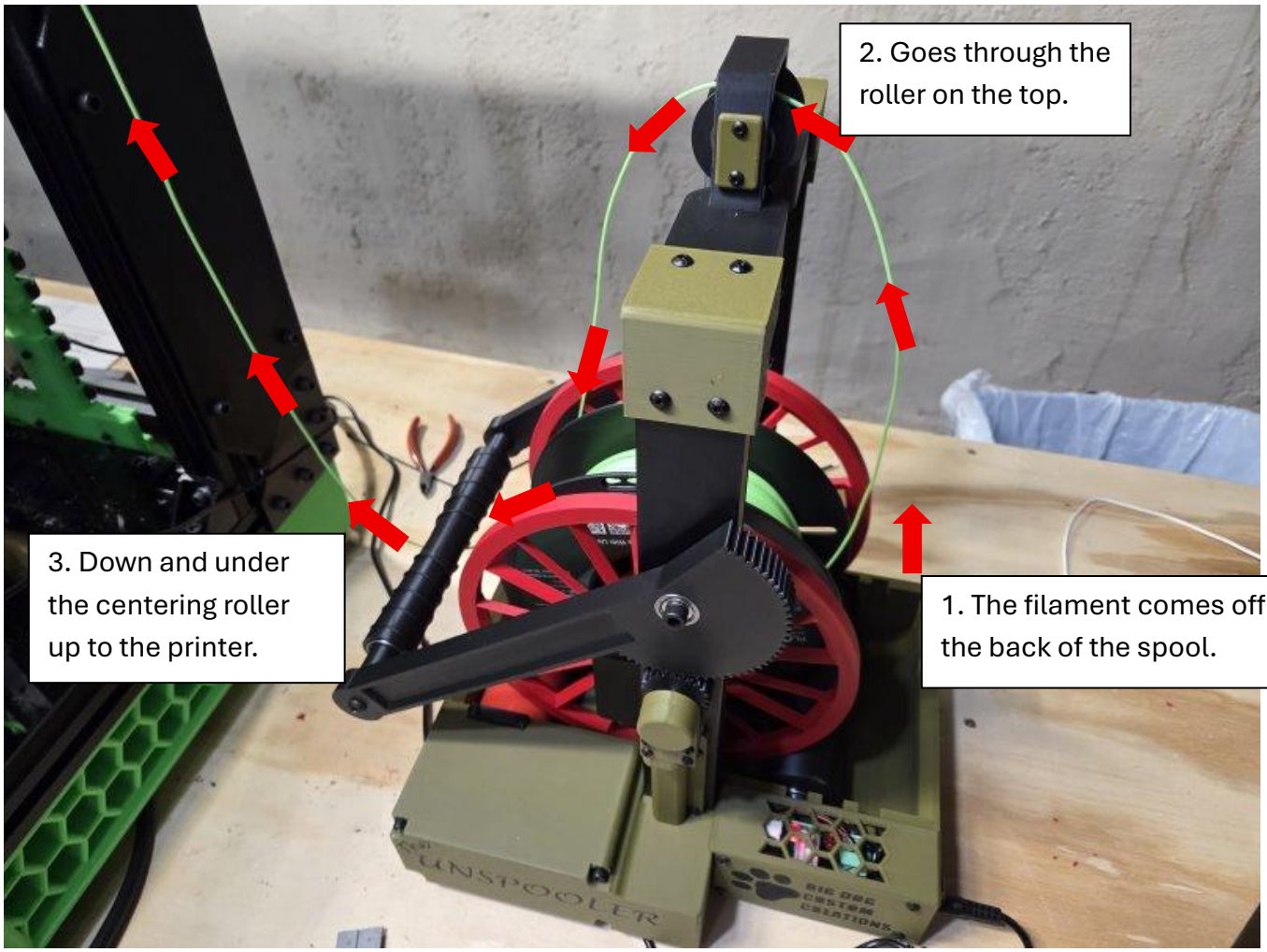
- 1 - Bringsmart CCM6N DC PWM Motor Speed Controller 6V-30V Variable Speed Drive 6A Positive Inversion Mini Switch Controller (Speed Controller)

Install the 2 Corner Décor pieces with 8 – M4x8mm Button head screws.



STEP 10 … UnSpooler Setup

UnSpooler Setup. Threading



Idler Roller Position



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